



The dark hand of the underworld: measuring the economic impact of organized crime

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Abstract

This study examines how organized crime affects China's economic climate using a unique dataset from its 2018–2020 “Saohei Chu’e” campaign. It introduces a city-level indicator of organized crime presence in 182 cities at the prefectural level or higher. Our findings provide robust evidence that cities with increased organized crime presence exhibit a significantly worse economic climate. The analysis identifies three mechanisms through which organized crime harms the economic climate: suppressing corporate innovation, inducing government corruption, and thwarting individual entrepreneurship. Additionally, heterogeneity analysis finds that the destructive economic impact of the organized crime is more severe in China's north-east region but weaker in cities characterized by higher levels of economic development and openness, as well as higher employment rates. This paper makes three contributions. First, it provides new evidence on the economic effects of organized crime in the Chinese context. Second, it enhances previous research by utilizing more theoretically grounded and detailed city-level data. Third, it offers empirical support for China's recent shift toward normalized anti-gangsterism policies and practices.

Keywords Organized crime · Economy climate · Law and economics · China

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Introduction

Organized crimes have been demonstrated to exert substantial negative impacts on the local economy (Global Organized Crime Index 2023, [n.d.](#)). These criminal organizations, from the Italian Mafia and the Russian Bratva to the Hong Kong Triad and Japanese Yakuza, have often disrupted legitimate business activities through orchestrated violence, intimidation, and strategic alliances with corrupt authorities over sustained periods (Baradel, [2021](#); Calderoni, [2011](#); Lo, [2010](#); Tomass, [1998](#)). In contemporary society, they have even evolved beyond traditional predatory activities to engage in sophisticated business operations, generating far greater indirect economic losses through complex schemes of market manipulation and institutional infiltration (Dijk, [2007](#); Mirenda et al., [2022](#)). Therefore, countries severely affected by organized crime typically respond with exceptionally harsh enforcement measures to combat these criminal organizations (Paoli, [2007](#)).

China's remarkable economic growth since the 1980s has not been immune to this global challenge of organized crime. The four decades of rapid economic growth brought significant shifts in China's social structure. The sudden transition to a market economy led to widespread layoffs among state-owned enterprise employees, resulting in a substantial population of unemployed and displaced workers, as well as a fertile ground for organized crime groups (OCGs) to emerge (Uhm & Wong, [2021](#)). OCGs through the country have not only severely disrupted China's economic development but even posed a potential threat to the stability of its political system (Trevaskes, [2007](#)).

In response, the Chinese government has launched five campaign-style law enforcement against all OCGs (in 1983, 1996, 2001, 2010, and 2018). These campaigns have been widely recognized as effective, resulting in extensive prosecutions and convictions of OCG members (Yin & Mou, [2023](#)). However, unlike the rich literature on Italian Mafia that systematically examines the nexus between organized crime and regional economic development, existing research on Chinese organized crime has not sufficiently explored this crucial relationship. This research gap casts some doubt on the applicability of existing theoretical frameworks regarding organized crime-regional economy relationships to the Chinese context (Wang, [2019](#)).

A particularly challenging aspect comes from Gambetta's theory of private protection, which posits that non-state actors like the Mafia will step in to secure property rights through private protection services in the absence of effective state enforcement (Diego, [1993](#)). Based on this theory, some scholars argue that Chinese OCGs, emerged during the transition from a centrally planned to market economy like their Russian counterparts (Federico, [2001](#)), serve similar functions of private property rights protection and market operation maintenance, thereby potentially facilitating China's economic transition to some extent (Jiang & Wang, [2022](#)). This perspective aligns with the recent "Godfather Theory" emphasizing the role of gray-area intermediaries in maintaining social stability through peaceful dispute resolution (Chen et al., [2024](#)). On the other hand, extensive empirical

research has demonstrated that OCGs can generate significant negative economic impacts by decreasing local foreign direct investment (Daniele & Marani, 2011), creating industry barriers and excluding market competition (Ferrante et al., 2021; Slutzky & Zeume, 2023), causing economic losses equivalent to 16% of per capita GDP (Pinotti, 2015), and extracting up to 2.9% of GDP (NGO, n.d.). These conflicting theoretical perspectives and empirical findings necessitate a deeper examination of the Chinese context. Furthermore, the relationship between organized crime and economy in China becomes even more complex due to the intricate conceptual definitions of organized crime in the Chinese context and its distinct characteristics when compared to well-established criminal organizations such as the Italian Mafia and Japanese Yakuza (Wang, 2019).

Through analyzing prosecution data from China's most recent anti-organized crime campaign ("Saohei Chu'e") and corresponding economic climate indices, we posit that contemporary theories on the relationship between organized crime and economic development are fundamentally applicable to the Chinese context. Our findings suggest that excessive organized criminal presence adversely impact the Chinese economy through three potential mechanisms: stifling corporate innovation, luring corrupt officials, and suppressing residents' entrepreneurship.

Overall, by leveraging unique prosecutorial data from what may be China's final campaign-style anti-organized crime enforcement initiative, we contribute novel evidence and mechanistic insights to the literature on the economic impact of organized crime. Our analysis of the Chinese context expands the theoretical understanding of organized criminal activities' economic consequences. The policy recommendations derived from this study may inform strategies for mitigating organized crime-related activities both within China and in other jurisdictions.

Conceptualizing China's organized crime and "Saohei Chu'e" campaign

While the denotation of organized crime in China broadly aligns with international definitions, its connotation encompasses a broader spectrum of meanings. It extends beyond highly structured transnational Mafia-type organizations to include loosely connected smaller groups. This section will elaborate on how Chinese conceptualizations of organized crime correspond with commonly used terminology in international organized crime research, and introduce China's campaign-style anti-organized crime enforcement during 2018 to 2020 (the Saohei Chu'e" campaign).

China's organized crime

The concept of organized crime remains notoriously challenging to define. Although numerous scholars have engaged in theoretical debates on this topic (Decker & Pyrooz, 2014; Levi, 1998), James Finckenaue's work stands as a seminal contribution to the field. He astutely identifies violence, particularly violence aimed at

monopolizing markets for economic gain, as the most defining characteristic of organized crime (Finckenauer, 2005). Other distinctive attributes may encompass criminal sophistication, structural organization, self-identification, authority of reputation, operational scale, and continuity (Finckenauer, 2005). Based on these characteristics, OCGs manifest in various forms, but often structured around family ties, secret societies, cultural bonds, or economic alliances. Examples include the Italian Mafia, Hong Kong's Triads, Japan's Yakuza and South American cartels. These sophisticated criminal organizations are distinctly differentiated from street gangs, which typically feature informal and diffuse structures, public street-oriented activities, and expressive, non-specialized patterns of offending (Decker & Pyrooz, 2014).

The United Nations Convention against Transnational Organized Crime (2000), also known as the Palermo Convention, established the first internationally recognized definition of "organized criminal group" after synthesizing global consensus on combating organized crime. According to the Convention, an "organized criminal group" is defined as "a structured group of three or more persons, existing for a period of time and acting in concert with the aim of committing one or more serious crimes or offences established in accordance with this Convention, in order to obtain, directly or indirectly, a financial or other material benefit." It is evident that the international criteria for identifying organized crime largely align with academic perspectives, emphasizing four key elements: serious criminal activities, economic orientation, organizational structure, and temporal continuity. The international framework merely introduces a more specific numerical threshold of "three or more persons" in defining such organizations.

In China, organized criminal organizations have a long historical presence. Major social transformations—from feudal dynasties to colonial warlord conflicts, from the Kuomintang government to Communist Party rule, and through to the post-Mao era of Reform and Opening Up—have consistently provided substantial operational space for OCGs. Throughout Chinese history, various forms of OCGs have emerged, including secret societies (such as the Green Gang, Red Gang, Brotherhood, and Gelaohui), Heishehui (Tiandihui, Tianhui or Honglian, Hongmen), religious cults (Falun Gong), and more recently, transnational OCGs (Xia, 2008; Zhang & Chin, 2008). However, following multiple targeted law enforcement campaigns against these criminal organizations, most forms of OCGs have been effectively eliminated.

In current Chinese criminal justice practice, at least officially, only "criminal organizations of a Heishehui nature (COHN)" (Heishehui Xingzhi Zuzhi) are recognized as existing OCGs. This recognition is based on a conceptual framework within Chinese criminal justice, which regards Heishehui as an advanced form of OCG, somewhat analogous to mature criminal organizations like the Italian Mafia. However, compared to Mafia-type entities, China's OCGs are generally considered less sophisticated in terms of organizational structure and political penetration, typically maintaining connections only with mid- to low-level officials (Wang, 2017).

As a result, the definition of organized crime in Chinese criminal justice does not encompass terms like Heishehui or "Chinese Mafia," although these terms are used in academic discourse (Wang, 2017; Xia, 2008). According to Chinese criminal law, the typical form of organized crime, i.e., the COHN, are characterized by four core elements: (1) a relatively stable organizational structure and substantial membership

(typically interpreted as involving more than ten members); (2) acquisition of significant economic benefits through organized criminal activities (generally not less than 30,000 USD); (3) multiple instances of violence or threats of violence; and (4) control or influence over specific territories or industries through criminal activities or bribery, resulting in serious impact on economic order and social stability.

Another significant concept in China's organized crime conceptual framework is "E'shili" (Evil Force). Given the relatively stringent criteria for identifying COHN, the Chinese criminal justice has developed the supplementary concept of "E'shili" to address groups that fall below this threshold. Evil forces are defined as groups that regularly congregate (typically consisting of three to ten persons) and engage in multiple instances of illegal activities within specific territories or industries through violence, threats, or other means. These groups engage in malicious activities, oppress the masses, and disrupt social and economic order, creating significant negative social impact, yet do not fully meet the criteria of COHN. Within the Chinese criminal justice framework, Evil forces are broadly understood as OCGs whose organizational structure and social harm have not reached the level of COHN, and their activities are not necessarily economically motivated. This concept roughly corresponds to the international notion of street gangs, except that they have largely retreated from overt street-level criminal activities due to the proliferation of surveillance cameras in China (Wang, 2019).

In brief, organized crime in China primarily encompasses two categories: COGN ("Heishili") and Evil Forces ("E'shili"). In Chinese law enforcement against organized crime, both groups are targeted simultaneously. Therefore, China's anti-organized crime campaigns are commonly abbreviated as "Saohei Chu'e" (literally, "sweeping away Heishili and eliminating E'shili").

China's 2018 "Saohei Chu'e" campaign

China has a long history of combating organized crimes. Prior to 2018, there were four nationwide campaigns against organized crimes in 1983, 1996, 2001, and 2010 (Xie, 2013). Similar to previous efforts, the 2018 campaign was launched in response to the recent surge in organized crime rates and several high-profile organized criminal activities that shocked the nation. However, what distinguishes this campaign is its unprecedented scale and depth. The campaign mobilized an unparalleled 24 agencies of the Communist Party of China ("CPC") and the government and ensured extensive grassroots mobilization, with the leadership pledging to eradicate all forms of organized crime (Yin & Mou, 2023). In terms of arrests, China's 2018 anti-organized campaign represents one of the most extensive in both Chinese and global history.

On January 24, 2018, the CPC and the State Council issued a directive for a three-year special campaign against all OCGs in China (The Central Committee of the Communist Party of China (CPC) and the State Council issued the Notice on the Special Campaign to Sweep away Black Societies and Eradicating Evil Forces 2018), which was promptly supported by the Supreme People's Court, the Supreme People's Procuratorate, the Ministry of Public Security, and the Ministry of Justice

on February 5, 2018 (Four Departments Jointly Issue a Notice on Severely Cracking Down on Organized Crime & Illegal Activities in Accordance with the Law, 2018). Empirical research indicates that the Chinese criminal justice system's response to organized crimes became significantly stricter during the campaign, substantiating the campaign's overall effectiveness (Feng & Zeng, 2024).

To dig out the maximum number of OCG members, the campaign primarily employed three strategies: encouraging voluntary surrender, conducting high-level inspections, and mobilizing grassroots judiciaries. First, rather than an abrupt commencement on January 24, 2018, the four responsible departments offered OCG members the opportunity to surrender voluntarily in exchange for lenient treatment during the first one month, while those who failed to comply would face heightened sanctions. Second, the central government dispatched high-level inspectors to different provinces in three rounds to oversee local anti-organized crime campaigns. The effectiveness of this measure was evident in both the judicial system and local markets (Feng & Zeng, 2024; Li & Cheng, 2024). Third, and probably the most effective, the anti-organized crime efforts were incorporated as a key metric in the performance evaluations of local law enforcement and judicial personnel, incentivizing them to prioritize and actively engage in anti-organized crime efforts (Xu & Liu, 2022). As a result, some procuratorates specially noted in their annual work reports that the number of prosecuted cases related to the "Saohei Chu'e" campaign ranks among the top in the province, promoting these achievements as professional accomplishments.

Under the impetus of various measures over a three-year period, China's anti-organized crime campaign achieved significant results with 3,644 COGN and 11,675 "evil forces" dismantled, approximately 230,000 suspects prosecuted, and over 43,000 individuals surrendering voluntarily (Zhang, 2021). More importantly, the subsequent sharp decline in prosecution rates—falling to less than 30% of 2020 levels by 2021 and maintaining this lower threshold thereafter (Supreme People's Procuratorate, 2022)—suggests the three-year campaign's effectiveness in identifying and dismantling the majority of OCG members across China. The national prosecutorial reports from this period therefore provide an excellent dataset for analyzing the relationship between organized crime and regional economy in China.

Empirical strategies

To provide empirical support for the impact of OCGs on China's economy, we employ economic climate indices as a measure of regional economic conditions, rather than traditional economic indicators such as GDP. This choice is motivated by several considerations. First, GDP only focuses on monetary benefits from economic activities, whereas economic climate indices offer a more comprehensive measurement framework, encompassing multiple dimensions including investment willingness, judicial fairness, government efficiency, and other indicators that better capture the overall economic impact of organized crime. Second, since organized crime primarily influences regional economies through its impact on business operations and decisions, these indices of economic climate—which directly reflect business perceptions and confidence—provide more precise

measurements of this mechanism. Third, and most importantly from a technical perspective, economic climate indices present fewer endogeneity concerns compared to traditional singular economic indicators that might be contaminated by criminal activities. Given these advantages, we argue that economic climate indices serve as a more appropriate measure for examining the OCGs' economic impact.

Our empirical analysis draws on micro unbalanced panel data from 194 prefecture-level and above cities in China for the years 2018–2020 and constructs a Fixed Effects model (*FE*) to estimate the impact of OCGs on the economy climate. The specific model setup is as follows:

$$EC_{\{i,t\}} = \alpha + \beta OCG_{\{i,t\}} + \Phi Controls + \gamma_{\{i\}} + \delta_{\{t\}} + \varepsilon_{\{i,t\}} \quad (1)$$

wherein, i denotes the prefecture-level and above city, and t represents the year. The dependent variable represents the economy climate, measured by the economy climate index from the “China City Economy Climate Database 2023.” The core explanatory variable $OCG_{\{i,t\}}$ represents the severity of OCG presence, measured by the ratio of the number of people prosecuted for suspected OCG-type crimes (including the COGN and evil forces) to the year-end permanent population. *Controls* is a set of control variables that include various characteristics of prefecture-level and above cities, considered over the “city-year” dimension. These variables aim to account for additional factors that may affect the city's economy climate and are not addressed by fixed effects. We mainly considered the potential impacts on the economy climate from aspects such as industrial structure, employment situation, trade structure, investment structure, level of financial development, investment in education and scientific research, level of property rights protection, and government service capability. The embodied control variables include the proportion of primary industry to GDP (*IND*), unemployment rate (*UNEMPLOY*), the natural logarithm of employee compensation (*LNWAGE*), the proportion of foreign trade to GDP (*TD*), the proportion of actual foreign investment to GDP (*FDI*), the proportion of the year-end balance of loans from financial institutions to GDP (*CREDIT*), the ratio of local fiscal expenditures on education and scientific affairs to total expenditures within the local budget (*EDURD*), the ratio formed by the number of patent infringement dispute judgements to the number of patent grants (*PP*), government efficiency (*GE*), and government care (*GC*). Detailed indicator construction and data sources are provided in Sect. 3.2. $\gamma_{\{i\}}$ represents city fixed effects, controlling for characteristic differences between prefecture-level and above cities. $\delta_{\{t\}}$ represents year fixed effects, controlling for macro variables that change over time. ε denotes the error term.

We are particularly interested in the coefficient β . A positive β indicates that cities with increased OCG presence have a better economy climate. A negative β aligns with the theory, as observed in other countries, of the “negative economic effects of OCG presence,” namely, that the greater the OCG presence, the worse the city's economy climate will become.

To enhance the robustness of our conclusion, in addition to employing a Fixed Effects model to estimate the impact of the OCGs on the economy climate, we

also applied a Random Effects model (*RE*) and a Pooled Ordinary Least Squares estimation model (*POLS*) to avoid estimation biases caused by model selection.

To ascertain the causal mechanisms through which OCGs affect the economy climate, we construct interaction terms for further analysis. The chosen moderating variables, significantly relevant to the economy climate, are identified based on established theories in organized crime economics. These variables, namely corporate innovation, corruption degree, and individual entrepreneurship, respectively cover the aforementioned three channels of corporate, corruption, and individual.

Additionally, considering that the impact of OCGs on the economy climate might exhibit regional heterogeneity, we categorized our sample based on multiple criteria such as geographic location, economic development level, economic openness level, and employment situation. On this basis, we introduced group variables and the interaction terms between group variables and the core explanatory variable in our baseline regression, to investigate the differences in the impact of OCGs on the economy climate across different regions.

Data and variables

Data source

The data for our research were sourced from several key databases, which include work reports of procuratorates in Chinese prefecture-level cities and above, statistical yearbooks of each province, the “China City Statistical Yearbook,” and the “China City Economy Climate Database 2023.” These datasets provide comprehensive information on the number of individuals prosecuted for suspected OCG-related crimes, the permanent resident population, economy climate indices, and other pertinent economic indicators for each city.

Information on the prosecution of suspected OCG-related crimes in each city is detailed in the work reports of procuratorates from 290 prefecture-level cities and above across China, spanning from 2018 to 2020. We manually collected and compiled the numbers of individual prosecuted from these reports and verified this data against the defendant numbers involving OCG-related crimes recorded by courts during the same period. After rectifying 68 instances of missing or conflicting data, we compiled a dataset of OCG-related prosecuted suspects encompassing 802 data points. The missing rates for the years 2018, 2019, and 2020 were 5.86%, 5.52%, and 12.07% respectively, reflecting a generally robust data structure.

The economy climate data was obtained from the “China City Economy Climate Database 2023” (Zhang et al., 2023). This database comprises 4 first-level indicators, 15 s-level indicators, and 24 third-level indicators, offering a quantitative assessment of the economy climate in all Chinese cities at or above the prefecture level from 2017 to 2021. The assessment encompasses four major dimensions: market, government, legal, and cultural environments, which is recognized as an authoritative source for economy climate metrics in China.

Additionally, the statistical yearbooks of each province and the “China City Statistical Yearbook” provide a diverse range of statistical indicators at the prefecture-level city scale, covering economic, financial, population, education, and scientific research metrics. The control variables utilized in this study are primarily derived from these statistical yearbooks, with specific sources detailed in the variable construction section.

Variable construction

Measuring the severity of OCG presence We want to use the number of OCG-related prosecuted suspects to describe the severity of OCG presence within a specific region, reflecting the prevalence or potential control exerted by local OCGs. To quantify this, we use the number of individuals prosecuted for suspected OCG-related crimes per ten thousand residents in each city, which is considered to be the most reasonable method. First, the choice to use the number of prosecutions instead of convictions is substantiated by the criminal funnel theory, which posits that the criminal justice system operates like a funnel where crimes pass through structured stages of victim reporting, police investigation, prosecutor action, and court sentencing, with each stage filtering out certain cases (Mayeux, 2018). Consequently, data from later stages are more likely to be distorted, leading to potential underestimations of crime severity due to procedural delays, evidence issues, trial durations, and confidentiality constraints. Additionally, the shift from reactive to proactive investigations in this campaign (Yin & Mou, 2023) suggests that relying on victim reports and police filings might not accurately reflect the actual criminal activity, as public reporting may underrepresent, and police data may inflate, the true extent of OCG presence due to aggressive law enforcement tactics (Xu & Shi, 2019). China’s procuratorates, serving as legally neutral supervisory bodies, are tasked with ensuring the quality of case handling, which includes a focus on prosecuting OCG-related activities while also correcting wrongful cases (Lin & Wu, 2022).

We argue that the number of prosecutions more accurately reflects the extent of OCG presence rather than the intensity of anti-OCG efforts. From the data, the number of prosecutions can only represent the number of people involved in organized crime in a city. To assert that this reflects the intensity of anti-OCG efforts would require further evidence for a secondary causal or logical inference. Furthermore, interpretation of prosecution data solely as a measure of enforcement intensity would imply significant regional discrepancies in enforcement, which is inconsistent with the uniform application of campaign-style law enforcement driven by central and local authorities. Besides, in contrast to some international contexts where OCGs may infiltrate high levels of government or influence electoral outcomes (Acconcia et al., 2014; Feo & Luca, 2017), in China, such connections typically do not extend beyond low-level police interactions and do not affect the broader enforcement campaigns. Therefore, without additional evidence, the number of prosecutions should be seen as an indicator of the presence of OCG members rather than the vigor of law

enforcement actions. This methodological approach aligns with established practices in the study of corruption (Dong & Torgler, 2013; Glaeser & Saks, 2006).

Measuring the economy climate This paper utilizes the city economy climate index from the “China City Economy Climate Database 2023” to assess the economy climate in each city. This index systematically evaluates the economy climate across four dimensions: government, market, legal, and cultural environments. These four primary indicators are derived from the four aspects of economic climate specified in China’s 13th Five-Year Plan Guidelines, which are considered the most comprehensive primary metrics for assessing China’s economic climate (Sanbao et al., 2023). For the determination of secondary indicators, this evaluation framework incorporates standards from international peers and academia, drawing from prominent assessment systems including the Doing Business (2019), the Business Environment Ranking and Index (2014), China’s Marketization Index (2018), China’s Urban Business Environment Index (2021), and China’s Urban Government-Business Relations Index (2022). Through this synthesis, thirteen secondary indicators were established. On the whole, the “China City Economy Climate Database 2023” provides comprehensive data on the economy climate for 296 cities at the prefecture level and above in China, covering the period from 2017 to 2020.

Control variables Given that the core explanatory variables and the dependent variables in this study are “city-year” dimensional indicators, the introduction of city fixed effects and year fixed effects in the Fixed Effects model aims to control for the impacts specific to city entities and temporal changes. However, additional “city-year” dimensional factors might still influence the economy climate. To minimize estimation biases from omitted variables, we incorporate the following control variables into our model: (1) Industrial structure (*IND*), measured by the proportion of primary industry to GDP, data sourced from China Stock Market & Accounting Research (CSMAR) database; (2) Employment conditions, specifically including the unemployment rate (*UNEMPLOY*) and the logarithm of staff compensation (*LNWAGE*), data sourced from CSMAR database; (3) Trade structure (*TD*), measured by the proportion of foreign economic trade to GDP, data sourced from the “China City Statistical Yearbook”; (4) Investment structure, measured by the proportion of actual foreign investment to GDP (*FDI*), data sourced from CSMAR database; (5) Level of financial development, measured by the ratio of the year-end balance of various loans by financial institutions to GDP (*CREDIT*), data sourced from CSMAR database; (6) Education and research investment level, measured by the ratio of local fiscal expenditure on education and science to total local fiscal budget expenditure (*EDURD*), data sourced from CSMAR database; (7) Property rights protection level, measured by the ratio of the number of patent infringement dispute judgments to the number of patent grants (*PP*), data sourced from China Judgments Online; (8) Government service capability, including government efficiency (*GE*) and government care (*GC*) indicators, data sourced from the “China Urban Politic-Commercial Relationship Report,” Economy Prediction System (EPS) database, and the “China Government E-service Capability Index Report.”

We utilized linear interpolation to address missing values in our indicators as comprehensively as possible and aligned these indicators across various databases by city name and year. Due to limitations in data availability and statistical discrepancies, we excluded samples with incomplete or mismatched data, ultimately compiling 509 observations. Specifically, we gathered data for 182 prefecture-level and above cities for each of the years 2018 and 2019, and 145 cities for 2020. We then conducted empirical estimations of our model using this final unbalanced panel dataset.

Empirical results and analysis

Baseline results

Table 1 presents the estimation results for Eq. (1) based on panel data from cities in China for the years 2018–2020. To enhance the robustness of the estimation results, we employed three different regression methods: Fixed Effects (*FE*), Random Effects (*RE*), and Pooled Ordinary Least Squares (*POLS*), with the results displayed in columns (1), (2), and (3) of Table 1, respectively. As shown in column (1) of Table 1, the Fixed Effects model estimates that the coefficient of the severity of OCG presence (*OCG*) is negative and significant at the 1% level, indicating that increased OCG presence will lead to worse city economy climate. The results in columns (2) and (3) of Table 1 also show that the estimated coefficient for the core explanatory variable, the severity of OCG presence (*OCG*), is significantly negative, corroborating the robustness of the results.

To intuitively show the regional variations in OCG presence, this paper reports the average number of prosecuted OCG members per 10,000 residents in each city over the years 2018 to 2020 in Fig. 1.

Robustness analysis

We conducted a series of tests to demonstrate the robustness of our main findings. First, we use the ratio of the number of prosecuted individuals suspected of OCG-related crimes to the number of people in public administration and social security as a proxy indicator of OCG presence (*OCG2*), and the regression results are shown in the column (1) of Table 2. The regression coefficient of *OCG2* is still significantly negative, indicating that the research conclusion of this paper is unaffected after using the alternative measurement of OCG presence.

Secondly, in order to overcome the endogeneity problem of using the original measure of *OCG*, we use the average value of *OCG* and *OCG2* of other cities in the same province as the instrumental variables of local *OCG* and *OCG2*. In general, the development conditions and policy strictness of cities within the same province are relatively similar. Therefore, it can be considered that there is a strong correlation between the average level of *OCG* in other cities within the same province and the local *OCG*. Furthermore, OCGs have a strong sense of territorialism, so

Table 1 Baseline regression results

	Dependent variable: city economy climate index (<i>EC</i>)		
	(1)	(2)	(3)
	FE	RE	POLS
<i>OCG</i>	-1.2381*** (0.4418)	-1.7345*** (0.3787)	-1.2381** (0.5651)
<i>IND</i>	-0.3196** (0.1394)	-0.1254*** (0.0332)	-0.3196* (0.1783)
<i>UNEMPLOY</i>	-2.0436 (9.6683)	-14.6565*** (2.2559)	-2.0436 (12.3681)
<i>FDI</i>	-56.8617 (38.3208)	9.7822 (38.5254)	-56.8617 (49.0219)
<i>CREDIT</i>	2.9262** (1.4799)	-0.4488 (0.4155)	2.9262 (1.8932)
<i>EDURD</i>	-24.2679 (21.4554)	12.5341** (5.5694)	-24.2679 (27.4468)
<i>PP</i>	0.0263 (0.0285)	0.0616*** (0.0180)	0.0263 (0.0365)
<i>GC</i>	-0.0319** (0.0127)	0.0384*** (0.0108)	-0.0319** (0.0162)
<i>LNWAGE</i>	-4.6983 (3.3682)	3.9791*** (1.2788)	-4.6983 (4.3088)
<i>GE</i>	-0.0376 (0.0266)	0.1014*** (0.0229)	-0.0376 (0.0340)
<i>TD</i>	-0.0151 (0.0282)	0.0137 (0.0170)	-0.0151 (0.0361)
<i>Constant</i>	101.0728*** (38.2288)	-2.6969 (14.2637)	96.9363** (47.4205)
City fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Adjusted R ²	0.79	0.62	0.87
Observations	490	509	509

Robust standard errors clustered at the city level are in parentheses. * indicates significance at the 10% level, ** indicates significance at the 5% level, and *** indicates significance at the 1% level. To mitigate reverse causality issues, explanatory variables and all control variables are lagged by one period. Coefficients on Controls are omitted for brevity.

the average level of *OCG* in other cities within the same province does not directly affect the local economy climate, thus satisfying the exogeneity condition of instrumental variables.

From the results of column (2) and column (3) in Table 2, we can see that the *p*-value of the under-identification test is 0, and the *F*-value of the weak

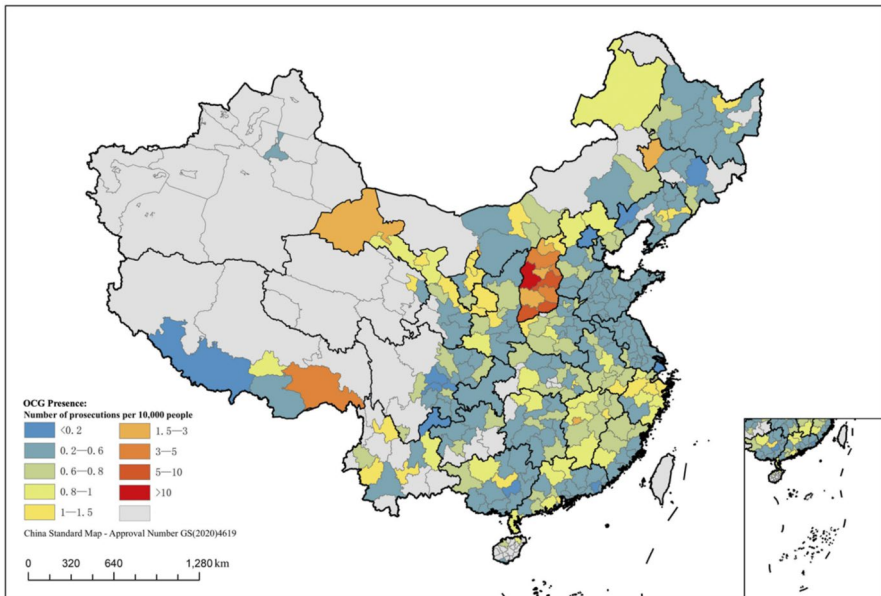


Fig. 1 The regional variations in OCG presence from 2018 to 2020

identification test is much higher than the critical value of 10% level, which also indicates the rationality of the selection of instrumental variables. In the results of the instrumental variable test, the coefficient of the explanatory variable is still significantly negative, which indicates the robustness of our findings.

Columns (4) and (5) respectively list the regression results excluding data in 2020 and the regression results with all variables winsorized at the 1% level, where the coefficients of the core explanatory variable remain significantly negative, indicating that our main finding still holds after excluding the influence of the COVID-19 pandemic and the influence of urban observation extremes.

Mechanisms

The baseline regression results indicate that cities with increased severity of OCG presence tend to have worse economy climate. To further explore the potential mechanisms through which OCG activities impact the regional economy climate, this paper builds upon the well-established theoretical analysis presented earlier and explores three potential mechanisms: innovation, corruption, and the public.

OCGs stifle innovation: the innovation repelling effect of OCG presence

Innovation is a primary driver of economic growth and a crucial indicator of the economy climate, but the OCG presence may repel and stifle innovation. From an

Table 2 Robustness test results

		Dependent variable: city economy climate index (<i>EC</i>)				
		(1)	(2)	(3)	(4)	(5)
		Alternative measure of OCG	IV-2SLS IV: OCG_ANO	IV-2SLS IV: OCG2_ANO	Excluded data of year 2020	All variables winsorized at 1% level
<i>OCG</i>			−2.9510*** (0.8992)		−0.9167* (0.5238)	
<i>OCG2</i>		−0.0140** (0.0067)		−0.0246** (0.0121)		
<i>OCG_win</i>						−1.3790*** (0.4468)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
UnderIdent-Chi (1) <i>P</i> value		0.00	0.00			
Kleibergen-Paap rk Wald F statistic		97.80	87.62			
Adjusted R ²	0.84				0.84	0.80
Observations	342	490	342	342	490	

Robust standard errors clustered at the city level are in parentheses. * indicates significance at the 10% level, ** indicates significance at the 5% level, and *** indicates significance at the 1% level. To mitigate reverse causality issues, explanatory variables and all control variables are lagged by one period. Coefficients on Controls are omitted for brevity

organizational perspective, OCGs are typically simple authoritarian entities composed of members with lower levels of education, based on violence, cultural or religious beliefs, and family ties (Campaniello et al., 2016; Catino, 2014). Such organizations are inherently averse to advanced knowledge and the accumulation of human capital (Caglayan et al., 2021). In terms of business operations, OCGs tend to favor labor-intensive industries that are easy to monopolize or launder money, and they are more willing to undertake government construction projects compared to investing in R&D-intensive industries (Ferrante et al., 2021; Moglie & Sorrenti, 2022). OCGs' disregard for human capital accumulation and market competition has been found to significantly suppress a city's innovation level (Caglayan et al., 2017, 2019).

To verify the hypothesis that innovation constitutes a significant mechanism through which OCG presence undermines the economy climate, we construct two variables to examine the local innovation level: the relative level of patents, measured by the number of patents granted per registered population (*PATENTRATE*), and the absolute number of patents, measured by the number of patents granted (*PATENTS*). The data used for calculating these indicators are sourced

from the “China City Statistical Yearbook.” We introduce the above two indicators and their interaction terms with the core explanatory variable *OCG* into the baseline regression and re-estimate the model.

The results are presented in columns (1) and (2) of Table 3. The table shows that the estimated coefficients for the interaction of the OCG presence with patents per capita ($OCGP \times PATENTRATE$) and with the number of patents ($OCGP \times PATENTS$) are both positive and significant at the 1% level. This indicates that in the case of increasing OCG presence, if the innovation level of a city declines more, then the economy climate will be more severely damaged. This finding supports the argument that OCGs may lead to the deterioration of the economy climate by stifling a city’s innovation.

OCGs lure corruption: the corruption inducing effect of OCG presence

Research has documented that OCGs often engage in corruption activities (Pezzi, 2019; Schneider, 2018), which enables them to disrupt the economy through multiple channels such as distorting resource allocation and deterring legitimate business

Table 3 Mechanism analysis: the innovation crowding-out effect of OCG presence

	Dependent variable: city economy climate index (<i>EC</i>)	
	(1)	(2)
<i>OCG</i>	−1.3779*** (0.4249)	−1.3854*** (0.4293)
<i>PATENTRATE</i>	0.1411*** (0.0310)	
$OCG \times PATENTRATE$	0.9282*** (0.3373)	
<i>PATENTS</i>		0.0001** (0.0000)
$OCG \times PATENTS$		0.0007*** (0.0002)
Control variables	Yes	Yes
City fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Adjusted R ²	0.80	0.80
Observations	490	476

Robust standard errors clustered at the city level are in parentheses. * indicates significance at the 10% level, ** indicates significance at the 5% level, and *** indicates significance at the 1% level. To mitigate reverse causality issues, explanatory variables and all control variables are lagged by one period. To alleviate the collinearity problem, variables involved in interaction terms, except for dummy variables, are mean-centered. Coefficients on Controls are omitted for brevity

(Abd Rashid et al., 2023; Ades & Tella, 1999; Dutta et al., 2017; Graycar & Villa, 2011; Mo, 2001). Blackburn et al. (Blackburn et al., 2017), through theoretical deduction, articulated how the existence of corruption serves as a prerequisite for the detrimental impact of OCGs on the economy. Following this line of inquiry, Neanidis et al. provided empirical evidence demonstrating that the involvement of corruption significantly amplifies or mitigates the economic devastation wrought by OCG activities (Neanidis et al., 2017). Therefore, we assume that when Chinese OCGs disrupt the economy climate, their actions are likely to be intertwined with the presence of corruption. We theoretically anticipate observing a significant moderating or enhancing effect of corruption on the damaging impact of OCG presence on the economy climate.

Integrating the methodologies of Glaeser & Saks and Chen & Li (Chen & Li, 2010; Glaeser & Saks, 2006), we consider it more rational to represent a region's corruption level by the number of officials under anti-corruption investigation (rather than the number of prosecutions and convictions) divided by the total number of local public officials.¹ We choose the five-year period from 2012 to 2016 as the time interval. This is because at the end of 2016, China suddenly launched a pilot reform of the national supervision system, which was a complete reshaping of China's previous anti-corruption path (Wu & Lin, 2024). Like Dong & Torgler (Dong & Torgler, 2013), we aim to avoid the influence of the new reform, which changed the corruption investigation agency and possibly its definition and standards of corruption-related activities in its starting years, in order to maintain the comparability of the corruption data.

Wang's "China Corruption Investigations Data" includes data on nearly 20,000 officials under anti-corruption investigation from 2012–2016, with comprehensive and publicly available data on each official's name, position, location, rank, and investigation timing (Wang, 2020). This is currently the most comprehensive and publicly available database on corruption investigations in China. We use Wang's data to calculate the percentage of the total number of officials under anti-corruption investigation from 2012 to 2016 out of the total number of local public officials in those five years, as a measure of the city's corruption level (*GC*). Based on the city's corruption level, we construct a group variable *GChigh*, which marks cities with corruption levels in the top 50% of the period as 1, otherwise 0. We add the group indicator and its interaction terms with the severity of OCG presence (*OCG*) into Eq. (1) and estimate the coefficients of the interaction term to examine the impact of government corruption on the microeconomic effect of OCG presence.

Table 4 lists the results of the above analysis, where the coefficient of the interaction term between the historical percentage of officials under anti-corruption investigation and the severity of OCG presence ($OCG \times GChigh$) is significantly positive. This indicates that if the percentage of officials under anti-corruption investigation

¹ This is attributable to the fact that the investigation data not only align with the criminal justice funnel theory but also because anti-corruption investigations in China are directly impact the political careers of officials. As a result, investigative agencies will not easily initiate anti-corruption probes against officials without solid evidence of corrupt activities.

Table 4 Mechanism analysis: the corruption inducing effect of OCG presence

	Dependent variable: city economy climate index (EC)
	(1)
<i>OCG</i>	-2.2214*** (0.5492)
<i>GChigh</i>	-1.5261 (2.3360)
<i>OCG</i> × <i>GChigh</i>	1.7220** (0.7100)
Control variables	Yes
City fixed effects	Yes
Year fixed effects	Yes
Adjusted R ²	0.80
Observations	490

Robust standard errors clustered at the city individual level are in parentheses. * indicates significance at the 10% level, ** indicates significance at the 5% level, and *** indicates significance at the 1% level. To mitigate reverse causality, explanatory variables and all control variables are lagged by one period. To alleviate the col-linearity problem, variables involved in interaction terms, except for dummy variables, are mean-centered. Coefficients on Controls are omitted for brevity

in the past five years is higher in a region, the destructive effect of the OCG presence on the economy climate is more likely to be mitigated. Our findings align with the theoretical conclusion and empirical test drawn by Blackburn and Neanidis et al. regarding the Italian Mafia (Blackburn et al., 2017), which suggests while the Mafia exerts a negative economic impact, the presence of corruption may lessen this effect. The moderating effect is significant at the 5% level. The moderating effect, significant at the 5% level, confirms the existence and crucial role of corruption in OCG presence dynamics. This finding supports our hypothesis that OCG-induced corruption serves as a precondition for undermining the economic climate. This finding highlights the necessity of jointly tackling both organized crime and corruption within the context of safeguarding the economy climate.

OCGs suppress entrepreneurship: the entrepreneurship thwarting effect of OCG presence

OCGs may negatively affect local residents' entrepreneurial endeavors. First, OCGs' monopolistic practices and market disruption increase the barriers to entry and the risks for potential entrepreneurs (Ferrante et al., 2021). When OCGs dominate certain industries, they may use violence, intimidation, or collusion with officials to prevent new competitors from entering the market. This creates an unfavorable environment for entrepreneurship and discourages residents from starting their

own businesses. Second, OCGs' extortion of protection fees and other illegal levies directly erodes the profitability of local businesses and reduces the incentives for entrepreneurship (Asmundo & Lisciandra, 2008). When a significant portion of the profits is extracted by OCGs, entrepreneurs may find it difficult to sustain their businesses or may be deterred from starting new ventures altogether. Moreover, the presence of OCGs and their associated criminal activities may create a climate of fear and uncertainty, which further discourages entrepreneurship (Liu et al., 2023). Therefore, entrepreneurship should be an important mechanism through which the OCGs affect the economy climate.

We assess local entrepreneurial intent from two perspectives: the number of startups and the number of brands. The startup count is derived from the "2019 China Household Finance Survey Data." Since this survey is conducted biennially, data for 2018 and 2020 are unavailable, we then use the number of household startups (*STARTUP*) in 2019 as a proxy variable. The brand count (*BRAND*) data comes from the National Intellectual Property Administration, measured by the "number of trademark registrations per registered population." Incorporating these two indicators and their respective interaction terms with the severity of OCG presence into

Table 5 Mechanism analysis: the entrepreneurship thwarting effect of OCG presence

	Dependent variable: city economy climate index (<i>EC</i>)	
	(1)	(2)
<i>OCG</i>	-2.3247*** (0.5510)	-1.1673*** (0.4422)
<i>OCG</i> × <i>STARTUP</i>	0.0302*** (0.0111)	
<i>BRAND</i>		0.2958* (0.1753)
<i>OCG</i> × <i>BRAND</i>		2.4887* (1.4581)
Control variables	Yes	Yes
City fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Adjusted R ²	0.76	0.80
Observations	452	490

Robust standard errors clustered at the city individual level are in parentheses. * indicates significance at the 10% level, ** indicates significance at the 5% level, and *** indicates significance at the 1% level. To mitigate reverse causality, explanatory variables and all control variables are lagged by one period. To alleviate the collinearity problem, variables involved in interaction terms, except for dummy variables, are mean-centered. Since that the variable *STARTUP* is constructed with the survey data in 2019, it is a city-dimension indicator and is absorbed by city fixed effects, making its coefficient unestimable and therefore not disclosed here. Coefficients on Controls are omitted for brevity

the regression, the results displayed in Table 5 show that all interaction term coefficients are significantly positive. This indicates that in regions with less suppressed entrepreneurial intent, OCG presence has a lighter detrimental effect on the economy climate. In particular, the number of new businesses significantly mitigates the destructive effect of the OCG presence at the 1% significance level, suggesting that OCG presence is most likely to damage the economy climate by suppressing the residents' willingness to start new enterprises.

Heterogeneity analysis

Given the pronounced socio-economic disparities and regional heterogeneity among Chinese cities, it is imperative to scrutinize the differential effects of OCG presence on the economy climate across varied areas. We categorized the sample into groups based on five aspects: geographical region, economic development level, investment openness, trade openness, and employment situation, and examined the differences in the microeconomic effects of OCG presence across these groups.

Specifically, in terms of geographical region, following the classification standards of the National Bureau of Statistics of China (National Bureau of Statistics, 2021), we divided the prefecture-level and above cities into the eastern, western, central, and northeastern regions, setting up three dummy variables: *Northeast*, *East*, and *Central*. For cities in the northeastern region, the *Northeast* variable is marked as 1, otherwise 0; for cities in the eastern region, the *East* variable is marked as 1, otherwise 0; for cities in the central region, the *Central* variable is marked as 1, otherwise 0. The regression results incorporating geographical region grouping variables and their interaction with the severity of OCG presence are listed in column (1) of Table 5, where the coefficient of the interaction term $OCG \times Northeast$ is significantly negative, indicating that the destructive effect of the OCG presence on the economy climate is more pronounced in the northeastern region.

Regarding the level of economic development, we grouped cities based on per capita GDP. If a city's per capita GDP ranks in the bottom 20% among all cities, the *GDPPClow* is marked as 1, otherwise 0; if a city's per capita GDP ranks in the top 20%, the *GDPPChigh* is marked as 1, otherwise 0. The regression results introducing the economic development level grouping variables and their interaction with the severity of OCG presence are listed in column (2) of Table 5, where the coefficient of the interaction term $OCG \times GDPPChigh$ is significantly positive, suggesting that in regions with higher levels of economic development, the negative impact of the OCG presence on the economy climate is weaker.

In terms of the degree of investment openness, we categorized cities based on the amount of actual foreign investment; if a city's actual foreign investment ranks in the bottom 20% among all cities, it is marked with *FDIRlow* as 1, otherwise 0; if a city's actual foreign investment ranks in the top 20%, *FDIRhigh* is marked as 1, otherwise 0. Regarding the degree of trade openness, we grouped cities based on the total amount of imports and exports. If a city's total imports and exports rank in the bottom 20% among all cities, *EXIMlow* is marked as 1, otherwise 0; if a city's total imports and exports rank in the top 20%, *EXIMhigh* is marked as 1, otherwise 0. The

regression results introducing grouping variables for the degree of investment openness and trade openness, along with their respective interactions with the severity of OCG presence, are listed in columns (3) and (4) of Table 5. Here, the coefficients for the interaction term $OCG \times FDIR_{high}$ and the interaction term $OCG \times EXIM_{high}$ are both significantly positive, indicating that in regions with higher degrees of economic openness, the negative impact of the OCG presence on the economy climate is weaker.

Regarding employment conditions, we categorized cities based on the unemployment rate. If a city's unemployment rate ranks in the bottom 20%, *Unemploylow* is marked as 1 among all cities, otherwise 0; if a city's unemployment rate ranks in the top 20%, *Unemployhigh* is marked as 1, otherwise 0. The regression results, after adding the aforementioned two grouping indicators and their interaction with the core explanatory variable—the severity of OCG presence—to the baseline regression, are presented in column (5) of Table 5. According to the table, the coefficient for the interaction term $OCG \times Unemploylow$ is significantly positive, indicating that in regions with greater employment conditions, the negative impact of the OCG presence on the economy climate is weaker (Table 6).

Conclusion

The negative effects of organized crime on the economy have been extensively documented, but the prevailing focus on the Italian Mafias may raise questions about the broader applicability of these findings and established theories. This paper investigates how Chinese OCGs impact the economy climate by leveraging city-level data during the country's "Saohei Chu'e" campaign from 2018–2020. Our findings provide robust evidence that increased severity of OCG presence leads to a worse economy climate. We propose three mechanisms through which OCGs harm the economy: 1) OCGs stifle corporate innovation, 2) OCGs induce government corruption, and 3) OCGs suppress individual entrepreneurship. Furthermore, heterogeneity analysis reveals that the destructive economic impact of OCG presence is more severe in China's northeast region, but more mitigated in cities with higher levels (top 20%) of economic development and openness, as well as lower unemployment rates (bottom 20%).

This study makes several contributions. First, it offers new evidence from the Chinese context on the economic consequences of OCG presence, enriching the global literature on organized crime and regional economy. Second, it employs granular city-level data, an improvement over prior studies using province-level statistics. Meanwhile, the introduction of criminological theories also advances the field's research to select more scientific data, such as prosecutorial indictments rather than trial conviction data. Third, it provides empirical support for China's recent transition to normalized anti-organized crime policies and the newly enacted Anti-organized Crime Law.

To conclude, our research underscores the importance of combating organized crime to foster a healthy economy climate, which may be achieved by providing favorable conditions for innovation and entrepreneurship, and combating

Table 6 Heterogeneity analysis

	Dependent variable: city economy climate index (<i>EC</i>)				
	(1)	(2)	(3)	(4)	(5)
	By Geo- graphical region	By economic development level	By investment openness	By trade openness	By employ- ment situation
<i>OCG</i>	-0.9404* (0.5442)	-1.5462*** (0.5890)	-2.0966*** (0.5986)	-2.3263*** (0.6589)	-1.7137*** (0.5724)
<i>Northeast</i> × <i>OCG</i>	-2.8394* (1.4965)				
<i>East</i> × <i>OCG</i>	-0.2028 (1.1449)				
<i>Central</i> × <i>OCG</i>	-0.2600 (0.9385)				
<i>GDPPC</i> _{low}		0.6138 (0.8373)			
<i>OCG</i> × <i>GDPPC</i> _{low}		0.2104 (0.8200)			
<i>GDPPC</i> _{high}		-1.4696 (1.1320)			
<i>OCG</i> × <i>GDPPC</i> _{high}		2.0510** (0.9897)			
<i>FDIR</i> _{low}			-0.1755 (0.7158)		
<i>OCG</i> × <i>FDIR</i> _{low}			1.1451 (0.7763)		
<i>FDIR</i> _{high}			-2.0488** (0.9314)		
<i>OCG</i> × <i>FDIR</i> _{high}			2.7502** (1.1826)		
<i>EXIM</i> _{low}				-0.8865 (0.9933)	
<i>OCG</i> × <i>EXIM</i> _{low}				1.0579 (0.7563)	
<i>EXIM</i> _{high}				-1.7064 (1.2256)	
<i>OCG</i> × <i>EXIM</i> _{high}				3.2843** (1.3246)	
<i>Unemploy</i> _{low}					-2.0828 (2.1064)
<i>OCG</i> × <i>Unemploy</i> _{low}					2.2171** (1.0928)
<i>Unemploy</i> _{high}					0.5206

Table 6 (continued)

	Dependent variable: city economy climate index (<i>EC</i>)				
	(1)	(2)	(3)	(4)	(5)
					(0.9419)
<i>OCG</i> × <i>Unemploy-high</i>					0.2997
					(0.9130)
Control variables	Yes	Yes	Yes	Yes	Yes
City fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.79	0.79	0.80	0.80	0.79
Observations	490	487	490	481	490

Robust standard errors clustered at the city individual level are in parentheses. * indicates significance at the 10% level, ** indicates significance at the 5% level, and *** indicates significance at the 1% level. To mitigate reverse causality, explanatory variables and all control variables are lagged by one period. To alleviate the collinearity problem, variables involved in interaction terms, except for dummy variables, are mean-centered. Since geographic location does not change over time, *Northeast*, *East*, and *Central* are city-dimension indicators and are absorbed by city fixed effects, making their coefficients unestimable and therefore not disclosed here. Coefficients on Controls are omitted for brevity

corruption, particularly in economically disadvantaged regions with lower economic openness and employment conditions. The insights gained can inform anti-organized crime initiatives not only in China but also in other countries grappling with the economic threats posed by OCGs.

Lastly, several limitations of this study should also be noted. First, Chinese government implemented unprecedented stringent “zero-COVID” policies in the final year (2020) of the “Saohei Chu’e” campaign (Dai et al., 2021; Zhengfa & Wanqiang, 2020), which decreased dramatically both criminal activities and law enforcement due to heavy restrictions on population mobility. Although technical adjustments have been made to the 2020 data, the pandemic’s influence on the result accuracy cannot be overlooked. Second, while the economy climate index provides a comprehensive measurement framework, it may not capture all aspects of economic impact from organized crime, particularly in informal sectors. Third, China’s unique conceptualization of organized crime may limit the generalizability of our findings to other institutional contexts.

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Data availability The datasets used and analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethical approval Not applicable.

Informed consent Not applicable.

Statement regarding research involving human participants and/or animals This research did not involve human participants or animals.

Competing interests The authors declare that they have no competing interests.

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